

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · \* = Δ-vs-frozen CI excludes 0  
9 metrics × 8 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie if within ε) · 10 K bootstrap

|                       | g0-frozen        | g0-pre-2X         | g0-pre-enc        | g0-pre-hd        | g0-s3DI-enc       | g0-s3DI-hd        | g0-sNoDI-enc      | g0-sNoDI-hd       | WINNER                              |
|-----------------------|------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------------------------|
| action_top1 ↑ better  | 38.356<br>±2.264 | 49.973<br>±2.301* | 50.027<br>±2.329* | 38.959<br>±2.219 | 49.260<br>±2.274* | 50.247<br>±2.301* | 49.644<br>±2.329* | 48.712<br>±2.301* | ViT-g-2.0 surgery<br>3stage_DI head |
| motion_cos ↑ better   | 0.008<br>±0.001  | 0.228<br>±0.006*  | 0.113<br>±0.004*  | 0.009<br>±0.001* | 0.121<br>±0.004*  | 0.119<br>±0.004*  | 0.124<br>±0.004*  | 0.126<br>±0.004*  | ViT-g-2.0 pretrain_2X<br>encoder    |
| taxonomy_f1 ↑ better  | N/A              | N/A               | N/A               | N/A              | N/A               | N/A               | N/A               | N/A               | tie                                 |
| future_mse ↓ better   | 1.645<br>±0.002  | 1.784<br>±0.003*  | 1.756<br>±0.002*  | 1.646<br>±0.002  | 1.718<br>±0.002*  | N/A               | 1.715<br>±0.002*  | N/A               | ViT-g-2.0 pretrain<br>head          |
| rollout ↓ better      | -0.008<br>±0.000 | -0.007<br>±0.000* | -0.011<br>±0.000* | -0.008<br>±0.000 | -0.006<br>±0.000* | N/A               | -0.007<br>±0.000* | N/A               | ViT-g-2.0 pretrain<br>encoder       |
| causal ↓ better       | 1.656<br>±0.002  | 1.776<br>±0.003*  | 1.744<br>±0.002*  | 1.656<br>±0.002  | 1.721<br>±0.002*  | N/A               | 1.719<br>±0.002*  | N/A               | ViT-g-2.0 pretrain<br>head          |
| tdist ↓ better        | 0.004<br>±0.001  | 0.005<br>±0.001*  | 0.001<br>±0.001*  | 0.004<br>±0.001  | 0.003<br>±0.001*  | N/A               | 0.004<br>±0.001   | N/A               | ViT-g-2.0 pretrain<br>encoder       |
| teacher_free ↓ better | 0.023<br>±0.000  | 0.032<br>±0.000*  | 0.025<br>±0.000*  | 0.023<br>±0.000  | 0.042<br>±0.001*  | N/A               | 0.039<br>±0.001*  | N/A               | ViT-g-2.0 pretrain<br>head          |
| maskratio ↓ better    | 0.052<br>±0.001  | 0.070<br>±0.001*  | 0.062<br>±0.001*  | 0.052<br>±0.001  | 0.060<br>±0.001*  | N/A               | 0.061<br>±0.001*  | N/A               | ViT-g-2.0 pretrain<br>head          |